RATNAJIT DHAR

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ratnajit-dhar | ratnajit-dhar | ratnajitdhar08

Chittagong - Bangladesh

SUMMARY

I am an undergraduate student in Computer Science & Engineering with a passion for continuous learning. As a competitive programmer, I've enhanced my logical and critical thinking skills. Currently, I'm focusing on Data Science & Machine Learning, aiming to contribute through innovative projects. My experience with various organizations has strengthened my leadership and organizational abilities, ensuring efficient task coordination and goal achievement.

EXPERIENCE

- **Green Lead** *June 2024 - Present*
  Tech Solutions Architect Remote
  - Worked on a project named Re-Energized Bangladesh on the Climate Intelligence Technology Team which involves creating a renewable energy dashboard in Excel for Bangladesh.
  - Currently developing a climate based chatbot using RAG.

EDUCATION

- **Chittagong University of Engineering & Technology** *March 2022 - Present*
  BSc. in Computer Science & Engineering
  - GPA: 3.83/4.00 (Till 6 semesters)

SKILLS

- **Programming Languages:** Python, C, C++
- **Database Systems:** MySql, Firebase
- **Frameworks & Libraries:** PyTorch, FastAPI, Flask, Streamlit, LangChain
- **Tools:** Github, Latex

PROJECTS

- **Green Lead Chatbot**
  Tools: Python, LangChain, Pinecone
  - Developed a custom chatbot for Green Lead capable of answering queries using organization-provided PDFs, website texts, and documents.
  - Integrated retrieval-based QA pipeline with document loaders, embeddings, and vector databases to ensure accurate responses.
- **Spam Classification with MLOps**
  Tools: Python,MLFlow, DVC
  - Built a spam classification system using machine learning with a complete MLOps workflow.
  - Implemented data ingestion, preprocessing, feature extraction and model training & evaluation.
  - Leveraged DVC data and pipeline versioning to ensure reproducibility and efficient collaboration.
  - Enabled experiment tracking and model version control to support scalable development and deployment using MLFlow
- **CGPA Calculator For CUET Students**
  Tools: Flask, Pandas, HTML, CSS, Google Colab
  - Developed a web application to automate CGPA calculation for CUET students.
  - Implemented a user-friendly interface allowing users to copy and paste grades for instant calculation.
  - Utilized Flask for backend development, ensuring smooth data processing.
- **Resume Screening App**
  Tools: Streamlit, Jupyter Notebook, Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn, Pickle, PyPDF2
  - Developed an automated resume screening system to optimize the recruitment process.
  - Implemented Natural Language Processing (NLP) techniques to analyze resumes, achieving improved categorization accuracy.
  - Created a user-friendly interface using Streamlit, ensuring ease of use for HR professionals.
  - Applied machine learning algorithms to predict suitable candidate categories based on resume content.

- **Customer Churn Prediction**
  Tools: Jupyter Notebook, Flask, Pandas, Scikit-learn, Matplotlib, Seaborn, Pickle, PyPDF2
  - Developed a predictive model to identify customer churn using ensemble machine learning techniques.
  - Leveraged Random Forest, XGBoost, and LightGBM to enhance prediction accuracy.
  - Implemented a Flask application for user-friendly model interaction and visualization.
- **Calories Burnt Prediction**
  Tools: Jupyter Notebook, FastAPI, Pandas, Scikit-learn, Seaborn, Pickle, HTML, CSS
  - Developed a machine learning model to predict calories burned during physical activities.
  - Achieved an R2 Score of 0.998 using XGBoost Regressor for high prediction accuracy.
  - Created a FastAPI application for real-time predictions and user interaction.
- **Handwritten Digit Recognition**
  Tools: Streamlit, TensorFlow, Keras
  - Developed a Streamlit app to recognize handwritten digits using the MNIST dataset.
  - Implemented TensorFlow and Keras for building and training the recognition model.
  - Created an interactive canvas for users to draw digits and receive predictions.
- **CUET Thesis & Project Portal**
  Tools: Firebase
  - Managed backend development for a centralized repository of student theses and projects.
  - Utilized Firebase for data storage and user authentication.
  - Designed the architecture to ensure easy access and sharing of academic work.
- **HR Analytics Dashboard using PowerBI**
  Tools: Power BI, Excel
  - Developed an HR analytics dashboard to analyze employee attrition factors.
  - Leveraged key HR metrics to provide insights into salary, job satisfaction, and demographics.
  - Created visualizations to facilitate data-driven decision-making for HR management.
- **TuitionEase**
  Tools: HTML, CSS, JavaScript, PHP, MySQL
  - Developed a comprehensive platform for effective tuition administration.
  - Designed user interfaces and managed backend functionalities to resolve scheduling conflicts.
  - Implemented database management using MySQL for efficient data handling.
- **Zenspace - Meditation Based Software**
  Tools: Java, MySQL
  - Developed a Java-based meditation application providing a serene digital environment.
  - Created an intuitive user interface using Java Swing for easy navigation.
  - Implemented features for diverse meditation sessions and secure donation processing.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL,S=IN SUBMISSION, T=THESIS

[C.2] **CUET-823@DravidianLangTech 2025: Shared Task on Multimodal Misogyny Meme Detection in Tamil Language**. In *Proceedings of the Fifth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages* https://aclanthology.org/2025.dravidianlangtech-1.57/

[C.1] **Short-Term Wind Speed Forecasting Using Different LSTM Deep Learning Methods**. In *Innovations in Electrical and Electronics Engineering. ICEEE 2024*, Lecture Notes in Electrical Engineering, vol 1295. Springer, Singapore. https://doi.org/10.1007/978-981-97-9112-5_12

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification** - Coursera-Stanford University — *March 2024*
- **Advanced Learning Algorithm** - Coursera-Stanford University — *June 2024*
- **ICPC Asia Dhaka Regional Contest 2024** - ICPC Foundation — *December 2024*
- **Data Science Methodology** - IBM Developer Skills Network — *March 2025*

Awards & Achievements

- **Champion - Team CUET_RSCube** — DU AI Challenge - National Datathon *2025*
- **Finalist - Top 20 - Team CUET_R2S4** — Bangladesh Blockchain Olympiad - AI Category *2025*
- **1st Runner-up** — Intra CUET ML Contest - IEEE CS CUET Student Branch Chapter *2025*
- **4th Place** — Shared Task on Misogyny Meme Detection - DravidianLangTech@NAACL *2025*
- **Honorable Mention** — ICPC Asia Dhaka Regional Contest *2024*
- **Specialist** — Codeforces *2024*
- **Academic Scholarship x 6** — Chittagong University of Engineering & Technology *2022-2025*
- **47th Place – Team CUET_StraightOuttaSRH** — CUET Inter University Programming Contest *2024*

Leadership Experience

- **Chair** *July 2025 - Present*
  IEEE Computer Society CUET Student Branch Chapter
- **Joint Organizing Secretary** *July 2025 - Present*
  CUET Computer Club
- **Joint General Secretary** *August 2025 - Present*
  Joyoddhoney
- **Vice President(Cultural)** *January 2025 - Present*
  Puja Udjapan Parishad, CUET
- **Head of Finance** *January 2025 - February 2025*
  Hult Prize @ CUET
- **General Secretary** *August 2017 - August 2018*
  Collegiates Cultural Club
- **Assistant Manager - Public Relations** *August 2021 - October 2022*
  Protyush

Volunteer Experience

- **Organizing Committee**
  Smarok 20 - Govt. Hazi Muhammad Mohsin College Rag Day Event
- **Organizing Committee - Assistant Secretary of Event Managament**
  Badhanhara 4.0 - Inter Departmental Cultural Competition by Joyoddhoney, CUET
- **Organizing Committee - Secretary of Finance**
  Badhanhara 5.0 - Inter Departmental Cultural Competition by Joyoddhoney, CUET
- **Organizing Committee**
  Boishakhi Adda - Bengali New Year Celebration by Collegiates Cultural Club
- **Organizing Committee**
  Chittagong Collegiate School Batch 18 Rag Day Event
- **Student Volunteer**
  Chittagong Collegiate School 180th year Celebration
- **Organizing Committee - Under Secretary General of Branding & Promotion**
  Chittagong Collegiate School Model United Nations